

Job stress, incivility, and counterproductive work behavior (CWB): the moderating role of negative affectivity[†]

LISA M. PENNEY^{1*} AND PAUL E. SPECTOR²

¹*University of Houston, Houston, Texas, U.S.A.*

²*University of South Florida, Tampa, Florida, U.S.A.*

Summary

The current study was designed to replicate findings from previous research regarding the relationships between job stressors, negative affectivity, and counterproductive work behavior (CWB) using peer-reported data and to assess the effects of workplace incivility on employee satisfaction and CWB. Results indicate that incivility, organizational constraints, and interpersonal conflict were negatively related to job satisfaction and positively related to CWB. Support was also found for the role of negative affectivity as a moderator of the relationship between job stressors and CWB, although only one significant moderator was found using peer-reported CWB. In general, the relationships between job stressors and CWB were stronger for individuals high in negative affectivity than for individuals low in negative affectivity. Copyright © 2005 John Wiley & Sons, Ltd.

Introduction

Counterproductive workplace behavior (CWB) has become an increasingly popular topic of study among organizational researchers. CWB refers to behavior by employees that harms an organization or its members (Spector & Fox, 2002) and includes acts such as theft, sabotage, verbal abuse, withholding of effort, lying, refusing to cooperate, and physical assault. Over the years, various researchers have studied a similar set of behaviors, though they have used different terminology depending on their theoretical focus, including: organizational delinquency (Hogan & Hogan,

*Correspondence to: Lisa M. Penney, University of Houston, Department of Psychology, 126 Heyne Building, Houston, TX 77204-5022, U.S.A. E-mail: lpenney@uh.edu

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1989), organization-motivated aggression (O'Leary-Kelly, Griffin, & Glew, 1996), organizational retaliatory behaviors (Skarlicki & Folger, 1997), workplace aggression (Baron & Neuman, 1996), workplace deviance, (Bennett & Robinson, 2000; Robinson & Bennett, 1995), revenge (Bies & Tripp, 1998), and antisocial behavior in organizations (Giacalone & Greenberg, 1997). To date, the majority of research in this area has focused on identifying environmental antecedents of CWB, such as job stressors, and identifying personality traits, such as negative affectivity, that may increase an individual's propensity to engage in CWB. However, while researchers agree on an interactionist perspective regarding the contributions of both person and environment variables in predicting behavior, few have investigated both with CWB in the same study (Aquino, Lewis, & Bradfield, 1999; Skarlicki, Folger, & Tesluk, 1999). Moreover, a number of studies in this area have examined the relationships between potential stressors and CWB using a single source of data, generally self-report. The purpose of the current study is twofold: (1) to assess the effects of a new job stressor, workplace incivility, on employee satisfaction, and CWB; and (2) to replicate findings from previous research regarding the relationships between job stressors, negative affectivity, and CWB using peer-reported data.

The reason for the growing interest in CWB is fairly obvious for CWB is, unfortunately, a common occurrence in organizations and can have a tremendous negative impact on both organizations in terms of lost productivity, increased insurance costs, lost or damaged property, and increased turnover (Baron & Neuman, 1996; Benminson, 1994; LeBlanc & Kelloway, 2002; Vigoda, 2002), as well as the people in them in terms of increased dissatisfaction (Keashly, Trott, & MacLean, 1994) and experienced job stress. One useful framework for understanding CWB comes from the job-stress literature. The model of the job-stress process presented by Spector (1998) asserts that environmental stressors are perceived by individuals as such, leading to the experience of negative emotions, such as anger or anxiety, which may be followed by reactions to the stressors, called job strains. Job strains can be classified as psychological, physical, or behavioral (Jex & Beehr, 1991). Behavioral strains are a means for individuals to cope with the stressor either by reducing the emotions elicited by the stressor (e.g., drinking alcohol, avoiding work) or by eliminating the stressor itself (e.g., talking with the supervisor, developing a solution). Behavioral strains such as yelling at a co-worker, staying home from work, and decreasing work quality or quantity can be considered CWB.

A number of job stressors have been linked to the performance of CWB, including role ambiguity, role conflict, workload, organizational constraints, and interpersonal conflict (Chen & Spector, 1992; Fox & Spector, 1999; Fox, Spector, & Miles, 2001; Miles, Borman, Spector, & Fox, 2002; Penney & Spector, 2002). Recently, Andersson and Pearson (1999) introduced a new construct, workplace incivility, which can also be considered a social job stressor. Workplace incivility refers to relatively mild, rude, and discourteous behavior in the workplace. Specifically, workplace incivility is defined as 'low-intensity deviant behavior with ambiguous intent to harm the target, in violation of workplace norms for mutual respect (Andersson & Pearson, 1999; p. 457).'

Conceptually speaking, workplace incivility overlaps a great deal with interactional justice. Interactional justice was originally conceptualized as a sub-dimension of procedural justice defined as the quality of interpersonal treatment received during the enactment of organizational procedures (Bies & Moag, 1986). Two of the five criteria for fairness in inter-personal treatment identified by Bies and Moag (1986) overlap with workplace incivility: (a) respect (e.g., courtesy, avoiding intentional rude or attacking behaviors); and (b) propriety of questions (e.g., avoiding improper questions). However, as Cortina, Magley, Williams, and Langhout (2001) point out, interactional justice focuses primarily on treatment received from a supervisor or others in authority as formal organizational policies and procedures are implemented, whereas workplace incivility is broader and includes treatment from any member of an organization and is not limited to formal procedural contexts.

The importance of examining the quality of inter-personal treatment outside of situations wherein formal organizational procedures are enacted was noted by Mikula, Petri, and Tanzer (1990). In their attempt to uncover the basic structure of unjust events, they asked participants in their study to describe occasions when they were unjustly treated and events that aroused their anger. Although participants' event descriptions included instances that clearly fit into the categories of distributive, procedural, and interactional justice, a large number of events referred to instances of poor inter-personal treatment such as inconsiderate, impolite, or aggressive conduct, treatment that violates a person's dignity, and acts that indicate lack of loyalty from others close. Mikula et al.'s results indicate that the quality of inter-personal treatment received at work is important regardless of whether it occurs during the execution of organizational procedures and suggests that a broader conceptualization of inter-personal treatment is needed. Workplace incivility as defined by Andersson and Pearson (1999) appears to fill this void.

As Andersson and Pearson (1999) point out, workplace incivility also overlaps with CWB, although CWB differs from incivility in several ways. First, CWB is a behavior that typically intends to inflict harm on a person or organization, whereas incivility is harmful behavior that is not necessarily intentional or malicious. That is, any harmful intent can be easily dismissed or denied. For example, the instigator may claim the behavior was due to ignorance or an oversight on his/her part, or may accuse the target of misunderstanding the behavior or being overly-sensitive (Andersson & Pearson, 1999; Cortina, Magley, Williams, & Langhout, 2001). Second, incivility includes behaviors that are milder than most forms of CWB. That is, while some indirect, passive forms of CWB (e.g., ignoring or making derogatory marks about someone) may also be considered incivility, openly hostile behaviors, such as making threats and sabotage, are not. By the same token, behaviors such as failing to invite someone to lunch would be considered workplace incivility, but not CWB. Finally, CWB and incivility differ in the direction from which they approach the social dynamic wherein these behaviors may occur. According to Spector's job stress model, incivility would be classified as a stressor, that is, some event or condition in the environment that requires a response, whereas CWB is considered a behavioral strain, or an individual's response to a perceived stressor. Thus, the CWB literature focuses on the actor and variables related to the performance of CWB, such as personality traits of actors and environmental conditions, including stressors, which may encourage or elicit CWB (e.g., lack of justice, organizational constraints). Workplace incivility, on the other hand, is primarily concerned with the target's perspective and reactions. That is, how does perceiving oneself to be the target or recipient of uncivil behavior affects attitudes and behavior. Thus, measures of CWB generally ask the participant to indicate how often they perform certain CWBs, while a measure of incivility would ask participants how often they have been the recipient of uncivil behavior. Of course, within a social exchange, a single behavior (e.g., refusing to help a co-worker) can be considered an act of CWB by one person and also be perceived as incivility by another. However, because we are examining these phenomena from an intra-individual perspective, it is important to make the distinction between *perceiving* oneself to be a target of incivility (i.e., a stressor) and *behaving* in response to some provocation or stressor (i.e., a behavioral strain), as the psychological experience is very different.

Andersson and Pearson (1999) acknowledged the social nature of workplace incivility and argued that acts of incivility have the potential to foster unpleasant exchanges or even lead to more serious behaviors. They describe an incivility spiral wherein an act of workplace incivility on the part of one individual leads to an act of incivility by a second party that may be of equal or increasing intensity. In the former case, the exchange is non-escalating. The latter case, on the other hand, results in an escalating spiral wherein each act is followed by an increasingly negative act. Situations such as these have the potential to lead to more intense forms of CWB, perhaps resulting ultimately in aggression or violence wherein the intent to inflict harm is indisputable.

Workplace incivility also has much in common with employee abuse (Keashly, Hunter, & Harvey, 1997; Keashly et al., 1994), mobbing/bullying (Leymann, 1990; Zapf, Knorz, & Kulla, 1996), social undermining (Duffy, Ganster, & Pagon, 2002), and inter-personal conflict (Spector & Jex, 1998). While these constructs share overlapping behavior sets with workplace incivility, they generally refer to patterns of behavior with clear hostile intent which occur over time. In contrast, workplace incivility is milder and is often ambiguous with regard to its underlying motive. Thus, the difference is largely a matter of degree. The unique contribution that is made by including incivility alongside these distinct, albeit related constructs is the idea that behaviors do not necessarily have to be clearly and deliberately hostile to negatively impact an individual or organization.

Because the topic of workplace incivility is relatively new, few studies have been conducted to examine the relationship between experienced incivility and target attitudes and behavior, such as job satisfaction and CWB (Pearson, Andersson, & Porath, 2000). Although not originally presented as a job stressor, workplace incivility does qualify as such. Thus, individuals who experience incivility could be expected to respond in much the same way as individuals who experience other job stressors.

Research has demonstrated that being exposed to job stressors such as organizational constraints and inter-personal conflicts is negatively related to job satisfaction (Chen & Spector, 1992; Spector, Dwyer, & Jex, 1988). Organizational constraints refer to situations at work that interfere with an individual's task performance, such as a lack of time, resources, or help from others. Interpersonal conflict refers to how well an individual gets along with others at work, such as the frequency of arguments and how often others are nasty (Spector & Jex, 1998). Being exposed to incivility would no doubt be an unpleasant experience as well. Therefore, the following hypotheses were proposed:

Hypothesis 1. The experience of job stressors will be negatively related to job satisfaction.

Hypothesis 1a: The experience of workplace incivility will be negatively correlated with job satisfaction.

Hypothesis 1b: The experience of organizational constraints will be negatively correlated with job satisfaction.

Hypothesis 1c: The experience of inter-personal conflict at work will be negatively related with job satisfaction.

Likewise, a number of studies have shown a relationship between experienced job stress and the performance of CWB (Chen & Spector, 1992; Miles et al., 2002; Penney & Spector, 2002). Andersson and Pearson's notion of an incivility spiral dovetails nicely with the job stress research, and suggests that the experience of incivility should be related to the performance of CWB as well.

Hypothesis 2. The experience of job stressors will be positively correlated with the performance of CWB.

Hypothesis 2a: The experience of workplace incivility will be positively correlated with the performance of CWB.

Hypothesis 2b: The experience of organizational constraints will be positively correlated with the performance of CWB.

Hypothesis 2c: The experience of interpersonal conflict at work will be positively correlated with the performance of CWB.

Many researchers have emphasized the importance of distinguishing between CWB that targets the organization (CWB-O) and CWB that targets other people (CWB-P) (Baron & Neuman, 1996; Bennett & Robinson, 2000; Fox et al., 2001; Goh, Bruursema, Fox, & Spector, 2003; Robinson & Bennett, 1995). Research has shown that certain variables have different relationships with the two categories of CWB. For example, Fox et al. (2001) reported that injustice, an organizational stressor, was more closely related with CWB-O, while conflict, an interpersonal stressor, was more closely related to CWB-P. Their finding lends support to the incivility spiral model which describes a social dynamic wherein a target of incivility directs his/her response to the perceived agent. Thus, individuals who experience incivility should be more likely to perform CWB that targets individuals as opposed to an organization. The same should also be true for interpersonal conflict.

Hypothesis 3. Experienced workplace incivility and interpersonal conflict will be more strongly correlated with CWB-P than CWB-O.

According to Andersson and Pearson (1999), whether or not an incivility spiral escalates depends upon several factors. One is that an escalating exchange is more likely to occur if the target experiences negative emotions. The role that experienced negative emotions play in the performance of CWB has been documented in a number of studies. For example, Spector and his colleagues (Chen & Spector, 1992; Fox & Spector, 1999; Fox et al. 2001) reported significant positive correlations between negative emotions such as frustration, anger, and anxiety and a variety of CWB including sabotage, interpersonal aggression, absenteeism, and theft. Moreover, the model of job stress presented earlier argues that negative emotions mediate the relationship between job stressors and CWB. In other words, the experience of negative emotions with its accompanying increase in physiological arousal energizes a behavioral response that may include CWB. These studies not only provide a theoretical explanation for the psychological mechanisms underlying the relationship between situations and behavior, but also hint at where in the process individual differences may become a factor. Thus, individuals whose personality predisposes them to experience negative emotions should be more likely to engage in CWB under stressful conditions.

Negative affectivity refers to the dispositional tendency to experience a variety of negative mood states (Watson & Clark, 1984). Individuals high in negative affectivity have been characterized as especially sensitive to minor frustrations and irritations, and are more likely to experience negative emotions, such as anxiety, guilt, anger, rejection, sadness, and distress. Researchers have suggested that the reason high-negative affectivity individuals experience more negative emotions is that they perceive the world more negatively than low-negative affectivity individuals (Chen & Spector, 1991; Jex & Beehr, 1991). When confronted with stressful conditions, including incivility wherein the intent to harm is open to interpretation, high-negative affectivity individuals may ascribe more malicious motives to the actor leading to increased negative emotional arousal, which may lead to CWB. Individuals low in negative affectivity, on the other hand, may give the actor the benefit of doubt and attribute the behavior to more benign causes, enabling them to proceed without feeling the need to respond.

At least three studies have already suggested that high-negative affectivity individuals are more likely to engage in CWB overall than low-negative affectivity individuals (Aquino et al., 1999; Douglas & Martinko, 2001; Skarlicki et al., 1999). Similarly, Fox and Spector (1999) and Fox et al. (2001) found that negative affectivity related to both CWB-O and CWB-P. Furthermore, in Fox et al. (2001) negative affectivity moderated the relation between constraints and conflict and CWB-P, but not CWB-O. Moreover, Skarlicki et al. (1999) found that, in addition to accounting for unique variance in CWB, negative affectivity moderated the relationship between perceived fairness and retaliation (a form of CWB). They reported that high-negative

affectivity individuals were more likely to engage in retaliation under conditions of low-perceived justice than low-negative affectivity individuals. Therefore, the following hypotheses were proposed:

Hypothesis 4: Negative affectivity will be positively related to CWB.

Hypothesis 5: Negative affectivity will moderate the relationship between experienced job stressors and the performance of CWB.

Hypothesis 5a: The relationship between experienced workplace incivility and the performance of CWB will be stronger for individuals high in negative affectivity than for individuals low in negative affectivity.

Hypothesis 5b: The relationship between experienced organizational constraints and the performance of CWB will be stronger for individuals high in negative affectivity than for individuals low in negative affectivity.

Hypothesis 5c: The relationship between experienced inter-personal conflict and the performance of CWB will be stronger for individuals high in negative affectivity than for individuals low in negative affectivity.

Current Study

With the exception of a few studies, the majority of CWB research has used a single source of data, generally self-report. Although self-report measures of CWB are easier to obtain, they also raise a few issues. First, self-report CWB questionnaires generally ask participants to report how often they engage in a variety of unseemly behaviors ranging from the annoying (e.g., failing to return a phone call, being late) to the unlawful (e.g., theft, physical assault). It is possible, therefore, that individuals interested in presenting themselves in a positive light may under-report the CWB they perform. Furthermore, employees may have greater motivation to under-report behaviors, such as theft that put them at risk of job loss or legal prosecution. Additionally, while a number of studies have examined the relationships between different person and situation variables and CWB, the majority of this research has used self-report questionnaires to obtain data for all variables. While self-report is undoubtedly the most efficient means of assessing individual personality, attitudes, and perceptions, the use of self-report as the sole source of data allows for the possibility that the observed relationships will be inflated due to participant biases common across measures.

One alternative to self-reported CWB is the use of hard criteria such as absences, suspensions, and warning letters. However, as Fox and Spector (1999) point out, these criteria often suffer from deficiency and contamination because, (a) they are unlikely to account for milder forms of CWB that do not merit formal action by the organization, such as working slowly or ignoring someone; (b) they are limited to instances where an employee is caught and formally disciplined by someone in authority; and (c) criteria such as absenteeism and tardiness may be a function of work or non-work issues unrelated to CWB. A more appropriate alternative is the use of peer-reported CWB. Having peers rate how often an employee engages in a variety of CWB should provide more comprehensive information than would be available from hard criterion data. Unlike supervisors who may have limited opportunity to directly observe employees' behavior on any given day, peers generally work more closely with employees, thus allowing them ample opportunity to observe a wider range of behavior. Moreover,

employees may be less likely to censor the behavior they display in front of their peers because the threat of punishment or other negative consequences is less than would be expected from a supervisor. Therefore, the second aim of this study was to replicate the findings of previous research examining the relationships between job stressors, negative affectivity, and CWB in a cross-sectional field study using both self- and peer-ratings of employee CWB.

Organizational Context

The respondents in this study were recruited mostly from night classes from a large public university in 2002. Because the university is located in a large urban area, a large number of individuals who enroll as students are working adults who want to complete their undergraduate education as opposed to individuals who enter college immediately following high school. The students at this university are older than typical undergraduate, with a mean age of 23. The ages of participants in this study ranged from 18 to 53 years, however 25 per cent of participants were over 25 years old and 10 per cent were over 30 years. The participants in this study were employed in a variety of jobs as opposed to a single organization, although information about the specific positions was not collected. Moreover, for many students at this university, the jobs they work are generally not temporary positions for the semester, but are permanent positions. Nearly half the participants in this study had worked in their jobs for at least two years, over 25 per cent had worked in their job at least three years, and 10 per cent had tenure in their jobs of nearly five years.

Method

Participants

Participants in this study were employed persons taking courses at a large public university in the Southeastern United States. Three hundred seven participants were surveyed from undergraduate night courses in computer science, interdisciplinary social science, management, engineering, chemistry, and psychology; although, three late afternoon (e.g., 3 or 4 pm) psychology courses were utilized as well. Participants' ages ranged from 18 to 53 years with an average of 23.3 years. Two hundred thirty-six participants were female (77 per cent) and seventy-one were male (23 per cent). On an average, participants were enrolled in 11.2 credit hours during the semester, they were surveyed. Nearly half (47 per cent) of participants worked at least 30 hours per week, and 53 per cent worked between 25 and 29 hours per week. Participants had been at their job for an average of two years, though the range was from 2 months to 15 years. In order to assure that participants had been on the job long enough to experience incivility and that their peers had sufficient time to observe their behavior, only data from participants who had been on the job for two months or more were included in the analyses, a total of 299.

Peer-responses were gathered by asking participants to give a questionnaire to one of their co-workers. The co-workers returned the questionnaire by mail and the response rate was 52 per cent resulting in 155 complete pairs. Co-workers reported working with participants for 2 to 126 months with an average of 21.4 months. Mean scores for each of the measures and demographic characteristics of participants with peer-data were compared to those without peer-data using *t*-tests for independent samples, and no significant differences were found.

Measures

Workplace incivility

Workplace incivility was assessed with a 43-item scale based on a number of existing measures (Workplace Incivility Scale (WIS), Cortina et al., 2001; Leymann Inventory of Psychological Terror (LIPT), Leymann, 1990; Workplace Aggression Research Questionnaire (WAR-Q), Neuman & Keashley, 2002). Items were chosen from existing measures of similar constructs (e.g., employee abuse, mobbing) based on how well they fit the definition of workplace incivility offered by Andersson and Pearson (1999). Four advanced industrial/organizational psychology graduate students were given a one-page description of workplace incivility along with the WIS, LIPT, and WAR-Q items, and were instructed to identify items that fit the workplace incivility construct. Items that were chosen by three or more students were included in the final workplace incivility measure. In the self-report version, participants were asked to indicate how frequently a supervisor or co-worker in their current job had subjected them to a variety of behaviors (e.g., ignored or excluded you from professional camaraderie, restricted your opportunities to speak, assigned degrading tasks to you). The version of the incivility measure presented to peers asked respondents to indicate how often they observed a situation wherein their co-worker was subjected to each behavior by someone at work. Items were presented in a five-point Likert scale ranging from 'never' to 'every day.' The incivility measure showed good internal consistency for both the self- and peer-report versions ($\alpha = 0.96$ and 0.97 , respectively).

Interpersonal conflict

The Interpersonal Conflict at Work Scale (ICAWS; Spector & Jex, 1998) is a four-item summated rating scale designed to assess the amount of conflict or discord an individual experiences at work. The ICAWS differs from the measure of incivility in that the items describe clearly hostile incidents and include two-way negative interactions (e.g., 'How often do you get into arguments with others at work?'). This is further illustrated by the moderate within and cross-source correlations with incivility (r 's = $0.23 - 0.59$), which also suggest that the conflict and incivility measures assess distinct constructs.

In the self-report version of the scale, respondents were asked how often they get into arguments at work and how often other people at work are rude to them, yell at them, and do nasty things to them. For the peer-report version of the scale, the items were re-worded to ask about the co-worker (participant). For example, the item 'How often are people rude to you at work?' was re-worded to 'How often are people rude to your co-worker at work?' Response options were presented in a five-point Likert format ranging from 'less than once per month or never' to 'several times per day' wherein higher scores indicate more conflict. Internal consistency reliability for the ICAWS in the current sample was 0.73 for self-report and 0.79 for peer-report.

Organizational constraints

The Organizational Constraints Scale (OCS) was developed by Spector and Jex (1998) to measure things or situations at work that interfere with task performance. Respondents were presented with 11 items, representing each of the 11 situational constraint areas identified by Peters and O'Connor (1980), and were asked how often they found it difficult or impossible to do their job because of each constraint. Five response-choices were given ranging from 'less than once per month or never' to 'several times per day.' The coefficient alpha reliability for the OCS in the current sample was 0.90 .

Negative affectivity

The ten-item negative affectivity scale of the Positive and Negative Affect Schedule (PANAS; Watson, Clark, & Tellegen, 1988) was used to assess negative affectivity. The construct validity of the negative affectivity scale has been demonstrated by its correlations with measures of psychological distress

(Watson et al., 1988), including the Hopkins Symptom Checklist (HSCL; Derogatis, 1974), the Beck Depression Inventory (BDI; Beck, Ward, Mendelson, Mock, & Erbaugh, 1961), and the State-Trait Anxiety Inventory State Anxiety Scale (A-State; Spielberger, Gorsuch, & Lushene, 1970). The negative affectivity scale of the PANAS consists of 10 words that describe negative emotions (e.g., afraid, scared, hostile). Participants were asked to indicate the extent to which they generally feel each emotion on a five-point Likert scale ranging from 'very slightly or not at all' to 'very much.' The coefficient alpha for the negative affectivity scale in the current study was 0.87.

Counterproductive workplace behaviors

The Counterproductive Work Behavior Checklist (CWB-C; Spector et al., 2004) was developed to measure a wide range of CWB by collapsing other available scales measuring these types of behaviors and eliminating overlapping items. Participants were presented with 45 items describing behavioral reactions and were asked to indicate how often they performed each behavior. The participants' peers were presented with the same items and were asked to indicate how often they observed the employee engaging in each behavior. The response choices were presented in a five-point scale ranging from 'never' to 'every day.' Higher scores indicate higher levels of CWB. The CWB-C demonstrated good internal consistency for both the self- and peer-report versions ($\alpha = 0.89$ and 0.97 , respectively). In addition to an overall score, the CWB-C also provides subscores for CWP-P, that is, behaviors that targeted individuals (e.g., stole something from a person at work; did something to make a person at work look bad; insulted someone about their job performance) and CWB-O or behaviors that target the organization (e.g., put in to be paid more hours than you worked; purposely did your work incorrectly; stole something belonging to your employer) (e.g., 'Tried to look busy while doing nothing.' 'Put in to be paid for more hours than you worked.').

Job satisfaction

A three-item measure developed by Cammann, Fichman, Jenkins, and Klesh (1979) was used to assess job satisfaction. The three items assess overall job satisfaction, as opposed to satisfaction with particular facets of the job (e.g., pay, workload) and one of the items was reverse-scored ('In general, I don't like my job'). Internal consistency reliability for the job satisfaction scale was 0.91.

Control variables

The demographic variables included in this study (gender, age, number of credit hours, and hours worked per week) were also used as control variables. Age and number of credit hours were measured as continuous variables, while gender (1 = male; 2 = female) and number of hours worked (1 = 25–29 hrs/wk; 2 = 30–34 hrs/wk; 3 = 35 + hrs/wk) were measured as categorical variables.

Procedure

Volunteers for the study were recruited during class time. Participants were informed that the study was being conducted as part of a research project, and that their participation was voluntary and could be terminated at any time without penalty. Only individuals who were currently employed were allowed to participate and in exchange for their time, participants were given refreshments (cookies), a thank-you note, and a small gift (a bookmark). Additionally, students in psychology courses received one extra-credit point to use toward their grade. Two other instructors, outside of psychology, also made extra-credit available to those who participated.

Participants were given a questionnaire containing all the self-report measures and an envelope containing a questionnaire for their peers. In order to protect the anonymity of the participants, they were instructed to generate their own secret code using any combination of letters or numbers of their choice

and to write the code on both questionnaires. Participants were then instructed to give the envelope to one of their co-workers. They were cautioned that this person should not be a supervisor, but someone they work with fairly closely. Participants were then asked to fill out their questionnaire and place it in a large box at the front of the classroom when finished. The envelope for the co-workers/peers contained the peer-report measures, instructions on how to complete the materials and return them to the researcher, a stamped envelope addressed to the researcher, and a thank-you note with a small gift (a bookmark). Additionally, the researcher's name, e-mail address, and phone number were included on the front of the envelope in the event that the participants or their peers had any questions.

Results

Descriptive statistics for the measures are presented in Table 1 and the inter-correlations among the measures and scale reliabilities are displayed in Table 2. All measures displayed good internal consistency with reliabilities ranging from 0.73 to 0.97. The distribution of scores on each of the measures appeared to be normal, with the exception of CWB, workplace incivility, and conflict. For these three measures, the distribution of scores was positively skewed. Convergence (i.e., a significant correlation) was found between self- and peer-ratings of workplace incivility ($r = 0.29$), interpersonal conflict ($r = 0.33$), CWB ($r = 0.24$), CWB-O ($r = 0.22$), and CWB-P ($r = 0.30$).

Support was found for the first hypothesis. Self- and peer-rated incivility were significantly correlated with job satisfaction (r 's = -0.46 & -0.38 , respectively), as were both self- and peer-rated conflict (r 's = -0.27 & -0.36 , respectively) and organizational constraints ($r = -0.36$). In general, support was also found for the second hypothesis. Self-reported incivility was positively correlated with self-reported CWB ($r = 0.47$), but not peer-reported CWB ($r = 0.11$, *ns*), while peer-rated incivility was significantly correlated with both self- and peer-rated CWB (r 's = 0.17 & 0.51 , respectively). Both self- and peer-reported conflict were positively correlated with self-reported CWB (r 's = 0.41 & 0.23) and peer-rated CWB (r 's = 0.22 & 0.50). Finally, organizational constraints were positively correlated with self-rated CWB ($r = 0.35$), but not peer-rated CWB ($r = 0.07$, *ns*).

The third hypothesis predicted that the correlations between conflict and incivility and CWB-P would be stronger than the correlations between conflict and incivility and CWB-O. The correlations were compared using the Hotelling-Williams *t*-test for dependent correlations, (Williams, 1959) and partial support was found. The correlations between interpersonal conflict and CWB-P were

Table 1. Descriptive statistics for measures

Scale	N	Mean	SD	Min.	Max.
Negative affectivity	299	20.55	6.60	10	50
Workplace incivility (self)	299	66.68	22.17	43	202
Workplace incivility (peer)	152	66.84	27.06	42	202
Interpersonal conflict (self)	299	5.21	1.84	4	16
Interpersonal conflict (peer)	155	5.83	2.64	4	17
Organizational constraints	299	20.13	8.31	11	55
Job satisfaction	299	12.92	4.08	3	18
CWB (self)	298	60.19	11.43	45	136
CWB (peer)	155	61.06	23.64	45	156
CWB-O (self)	298	31.30	6.78	21	69
CWB-O (peer)	155	30.08	11.05	21	72
CWB-P (self)	298	26.17	5.43	22	62
CWB-P (peer)	155	28.28	10.94	22	85

Table 2. Intercorrelation matrix with scale alpha coefficient reliabilities on the main diagonal

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Negative affectivity	(0.87)												
2. Incivility (self)	0.24**	(0.96)											
3. Incivility (peer)	0.08	0.29**	(0.97)										
4. Conflict (self)	0.26**	0.49**	0.23**	(0.73)									
5. Conflict (peer)	0.08	0.31**	0.59**	0.33**	(0.70)								
6. Constraints	0.13*	0.57**	0.21**	0.35**	0.23**	(0.90)							
7. Job satisfaction	-0.20**	-0.46**	-0.38**	-0.27**	-0.36**	(0.91)							
8. CWB (self)	0.30**	0.47**	0.17*	0.41**	0.23**	0.35**	(0.89)						
9. CWB (peer)	0.19*	0.11	0.51**	0.22**	0.50**	0.07	0.24**	(0.97)					
10. CWB-O (self)	0.29**	0.44**	0.19*	0.31**	0.20*	0.32**	0.91**	0.19*	(0.81)				
11. CWB-O (peer)	0.20*	0.10	0.45**	0.19*	0.39**	0.05	-0.25**	0.22**	0.95**	0.22**	(0.94)		
12. CWB-P (self)	0.25**	0.39**	0.14	0.45**	0.20*	0.28**	-0.20**	0.86**	0.24*	0.58**	0.16*	(0.86)	
13. CWB-P (peer)	0.16*	0.11	0.53**	0.24**	0.55**	0.07	-0.22**	0.23**	0.94**	0.13	0.79**	0.30**	(0.95)

* $p < 0.05$, ** $p < 0.01$; n 's = 299 for self-report, 155 for peer-report.

significantly greater than the correlations between conflict and CWB-O when both variables were either self-report [$t(295) = 2.98, p < 0.01$] or peer-report [$t(152) = 3.52, p < 0.01$]. However, no significant difference was found for incivility and CWB. Fourth hypothesis was also supported. Negative affectivity was significantly correlated with both self-reported CWB ($r = 0.30$) and peer-reported CWB ($r = 0.19$).

Fifth hypothesis predicted that negative affectivity would moderate the relationships between job stressors (workplace incivility, interpersonal conflict, and organizational constraints) and CWB. Specifically, it was expected that the relationship between job stressors and CWB would be stronger for individuals high in negative affectivity than for individuals low in negative affectivity. These relationships were tested using moderated regression, wherein the predictors, the moderator (negative affectivity), and their product terms were entered into the regression simultaneously. Significant beta weights for the product terms would indicate an interaction effect. Both self- and peer-reported CWB were used as criteria. Because peer data were not available for all participants, the sample size for the analyses using peer data was 155, whereas the sample size for the analyses using self-report data only was 299.

The regression analyses were also conducted by entering the demographic variables (gender, age, number of credit hours, and number of hours worked) as controls in the first step followed by the predictor, moderator, product term in the second step. Although age and gender had significant relationships with CWB in some of the analyses (i.e., males were more likely to report performing CWB and co-workers reported more CWB for younger employees), the moderator outcomes were not affected and thus, for the sake of clarity, those analyses are not reported here.

A significant interaction between self-reported incivility and negative affectivity was found when self-reported CWB was used as the criterion ($\beta = 0.81$; see Table 3). The interactions were plotted using the simple effects equations (Cohen, Cohen, West, & Aiken, 2003), using values one standard deviation above and below the mean. The pattern of the data showed that when negative affectivity was high, the line depicting the relationship between incivility and CWB had a steeper slope than when negative affectivity was low (see Figure 1). However, no significant interaction effect was found between peer-reported incivility and negative affectivity ($\beta = -0.13, ns$) for self-reported CWB. Moreover, no significant interaction effects were found when peer-reported CWB was used as the criterion (see Table 3).

Table 3. Moderated regression of CWB onto workplace incivility and NA

Criterion: CWB (Self-Report)	b	Standardized β coefficient	Adjusted R^2	ΔR^2
Incivility (self-report)	-0.04	-0.08	0.22	0.22
Negative affectivity	-0.52	-0.30	0.25	0.03
Workplace incivility (self-report) $\times$ Negative affectivity	0.01	0.81**	0.29**	0.04**
Workplace incivility (peer-report)	0.10	0.24	0.02	0.02
Negative affectivity	0.62	0.36 [†]	0.10	0.08
Workplace incivility (peer-report) $\times$ Negative affectivity	-0.02	-0.13	0.09	-0.01
Criterion: CWB (peer-report)				
Workplace incivility (self-report)	-0.17	-0.17	0.00	0.00
Negative affectivity	-0.18	-0.53	0.03	0.03
Workplace incivility (self-report) $\times$ Negative affectivity	0.12	0.38	0.03	0.00
Workplace incivility (peer-report)	0.26	0.32	0.26	0.26
Negative affectivity	0.77	0.02	0.28	0.02
Workplace incivility (peer-report) $\times$ Negative affectivity	0.07	0.24	0.28	0.00

[†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$; n 's = 299 for self-report, 155 for peer-report.

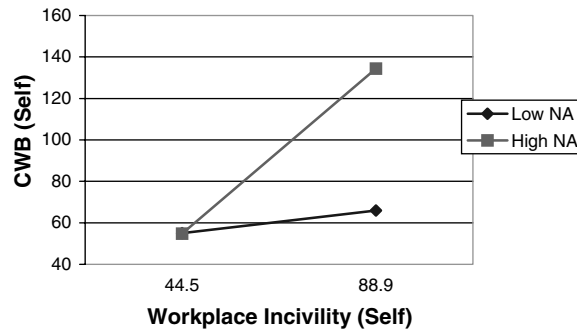


Figure 1. NA as a moderator of the relationship between workplace incivility and CWB

The outcome of the moderated regression for the negative affectivity-conflict interactions were more supportive than the negative affectivity-incivility results. The interaction between negative affectivity and self-reported conflict was significant against the self-report CWB criterion ($\beta = 0.76$), and the interaction between negative affectivity and peer-reported conflict was significant for the peer-report CWB criterion ($\beta = 0.61$; see Table 4). In each case the slopes of the regression lines depicting the relationship between conflict and CWB were steeper for individuals high in negative affectivity than individuals low in negative affectivity (see Figures 2 and 3). However, the interaction between negative affectivity and self-reported conflict was not significant against peer-reported CWB ($\beta = -0.67$, *ns*), and the interaction between negative affectivity and peer-reported conflict was not significant against self-reported CWB ($\beta = 0.64$, *ns*).

Table 4. Moderated regression of CWB onto interpersonal conflict and NA

Criterion: CWB (Self-Report)	b	Standardized β coefficient	Adjusted R^2	ΔR^2
Interpersonal conflict (self-report)	-0.99	-0.16	0.17	0.17
Negative affectivity	-0.34	-0.20	0.21	0.04
Interpersonal conflict (self-report) $\times$ Negative affectivity	0.13	0.76**	0.24	0.03**
Interpersonal conflict (peer-report)	0.21	0.05	0.04	0.04
Negative affectivity	0.32	0.19	0.12	0.08
Interpersonal conflict (peer-report) $\times$ Negative affectivity	0.03	0.19	0.12	0.00
Criterion: CWB (Peer-Report)				
Interpersonal conflict (Self-Report)	4.16	0.37	0.04	0.04
Negative affectivity	0.96	0.28	0.06	0.02
Interpersonal conflict (self-report) $\times$ Negative affectivity	-0.08	-0.24	0.06	0.00
Interpersonal conflict (peer-report)	0.18	0.02	0.24	0.24
Negative affectivity	-0.58	-0.17	0.26	0.02
Interpersonal conflict (peer-report) $\times$ Negative affectivity	0.18	0.61*	0.28	0.02*

[†] $p < 0.10$, * $p < 0.05$, ** $p < 0.01$; n 's = 299 for self-report, 155 for peer-report.

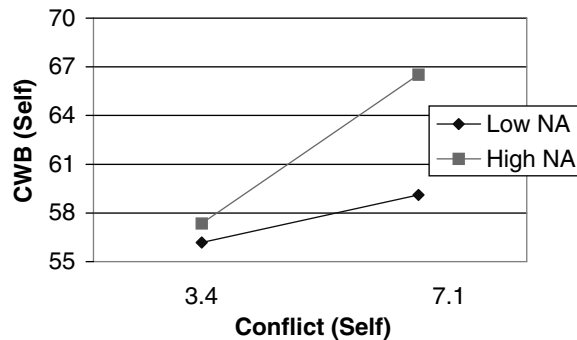


Figure 2. NA as a moderator of the relationship between interpersonal conflict and CWB (Self-ratings)

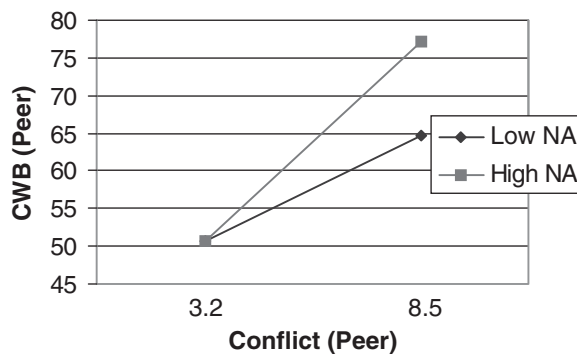


Figure 3. NA as a moderator of the relationship between interpersonal conflict and CWB (Peer-ratings)

Regarding organizational constraints and negative affectivity, the interaction was significant against self-reported CWB ($\beta = 0.53$). As predicted, a graph of the interaction indicated that the relationship between constraints and CWB was stronger for individuals high in negative affectivity than for individuals low in negative affectivity (see Figure 4). However, this finding was not replicated using peer-reported CWB ($\beta = 0.03$, *ns*; see Table 5).

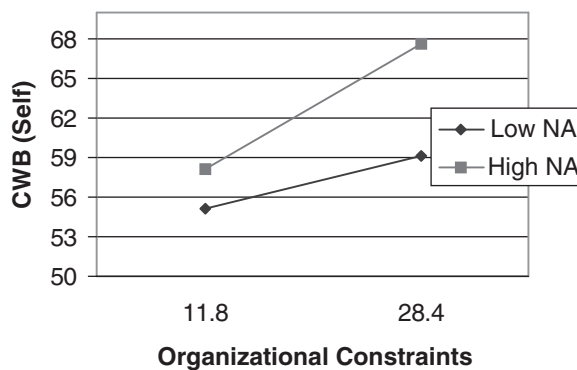


Figure 4. NA as a moderator of the relationship between organizational constraints and CWB

Table 5. Moderated regression of CWB onto job constraints and NA

	b	Standardized β coefficient	Adjusted R^2	ΔR^2
Criterion: CWB (Self-Report)				
Job constraints	-0.11	-0.08	0.12	0.12
Negative affectivity	-0.06	-0.04	0.18	0.06
Job constraints $\times$ Negative affectivity	0.03	0.53*	0.19	0.03*
Criterion: CWB (Peer-Report)				
Job constraints	0.09	0.04	0.00	0.00
Negative affectivity	0.60	0.18	0.03	0.03
Job constraints $\times$ Negative affectivity	0.00	0.03	0.02	-0.01

† $p < 0.10$; * $p < 0.05$; ** $p < 0.01$; n 's = 299 for self-report, 155 for peer-report.

Discussion

While a number of studies in the job-stress literature have demonstrated that trait negative affectivity and the experience of stressors on the job is related to job dissatisfaction and the performance of CWB, the majority of those studies used data from a single source (i.e., self-report). The purpose of the current study was to replicate those findings using multiple data sources (i.e., self- and peer-report) and to examine the relationship between a newly identified job stressor, workplace incivility, and employee satisfaction and CWB. In general, the results were supportive of the hypotheses, although the results for the moderator hypotheses were mixed.

One of the most consistent findings in the job stress literature is the negative relationship between experienced stressors and job satisfaction. In the current study, all of the correlations between job stressors (incivility, conflict, and constraints) and job satisfaction were significant and negative, regardless of whether the stressors were rated by the participants themselves or their peers. In fact, the correlations between both self- and peer-rated stressors and satisfaction were similar in magnitude suggesting that single-source biases did not likely inflate the correlations. Moreover, all of the correlations between peer-rated stressors and self-rated CWB and job satisfaction were significant. However, the results using peer-ratings of CWB were mixed. Self-rated incivility and constraints were not significantly related to peer-rated CWB; although, peer-rated CWB was significantly related to self-rated negative affectivity, conflict, and job satisfaction (negatively). While workplace incivility is qualitatively milder than other forms of mistreatment, these findings suggest that it may still negatively impact how people feel about their jobs. However, the possibility of the causal dynamics working in the opposite direction where dissatisfied individuals are more likely to be targeted for incivility could not be ruled out, given the design of this study. It is also possible that some other third variable, such as poor job performance, may lead to both experienced dissatisfaction and uncivil treatment from others at work.

One of the goals of this study was to replicate previous findings of the relationships with negative affectivity, job stressors, and CWB using peer data. With the exception of self-reported incivility and constraints, all other variables in the study were significantly correlated with both self- and peer-rated CWB. Thus the results from the existing literature, which used only self-report data, were generally replicated using peer data. Although the argument for using peer-ratings of CWB is that they may be less biased than self-ratings of CWB as individuals may under-report the CWB they perform to make themselves look good, it is also possible they could be more biased. If an impression-management bias is expected to lead to individuals reporting less CWB, would it not also lead to individuals displaying

less CWB in front of their co-workers? After all, the consequences for admitting to CWB in an anonymous survey would be far less than performing CWB in front of a peer who may not approve. In our sample, only 1 per cent of participants reported engaging in no CWB whatsoever, while 16 per cent of peers reported never observing their co-worker perform any CWB. Because it is unlikely that the participants were motivated to over-report the CWB they perform, this discrepancy suggests that peer-ratings of CWB may be a deficient indicator of actual CWB. While it is also possible that this finding may have been the result of participants giving the survey to co-workers who are also their friends and thus, motivated to make the participants look good as well, it nonetheless, raises the issue of just how unbiased or biased peer ratings of CWB really are. In the future, researchers may want to use randomly selected co-workers to provide peer ratings of CWB in order to address this issue more directly.

The results of the current study are consistent with other researches showing a link between job stressors and CWB, and provide support for Spector's (1998) model of job stress and CWB. Furthermore, they support Andersson and Pearson's (1999) notion of an incivility spiral and suggest that being the target of workplace incivility may increase an individual's likelihood of engaging in CWB. Although the causal direction of the relationship, that is, whether incivility leads to CWB or CWB leads to incivility, could not be determined from the research design, either alternative would be consistent with Andersson and Pearson's (1999) notion of an incivility spiral. An incivility spiral, as an interactive social process, affects and is affected by the behavior of the involved parties (target and actor/perpetrator), as well as the social context (Andersson & Pearson, 1999). This implies that being subjected to incivility could provoke an individual to engage in retaliatory acts, such as CWB. In other words, the target becomes an actor. Moreover, it also implies that individuals who engage in harmful acts, including CWB, would be the target of more workplace incivility. That is, the actor becomes a target. Examination of the workplace incivility-CWB relationship as an interactive ongoing process is an interesting idea, and one that can be expanded to include other stressors. After all, the performance of CWB by one individual could be experienced by others as incivility, conflict, or a job constraint which may increase their propensity to engage in CWB, thereby perpetuating the cycle. Although the current study did not attempt to capture the more complex aspects of the stressor-CWB relationship unfolding over time, the results suggest that this may be a promising area for future research.

Because Andersson and Pearson (1999) define an incivility spiral as a social interaction between individuals, it was expected that incivility and inter-personal conflict would be more strongly correlated with CWB-P than with CWB-O. The results were supportive of this trend for conflict, but not for incivility. One possible explanation is that victims of incivility must first perceive that the perpetrator performed the behavior with the intent to harm. In keeping with the definition of incivility as having an ambiguous intent to harm, some items from the incivility scale were open to interpretation. For example, an individual's supervisor may assign tasks far below an employee's skills, because the organization or the job itself constrains activities, rather than as an insult to the individual's abilities. Therefore, an individual who perceives organizational constraints as responsible for the incivility may target the organization instead of the person. Also, recent evidence suggests that the source of incivility (co-worker or supervisor) is related to the performance of CWB-P versus CWB-O. Lee and Spector (2004) found that employees who experienced conflict with a co-worker were more likely to perform CWB-P, but when the conflict was with a supervisor, they were more likely to perform CWB-O. Unfortunately, the current study did not examine the perceived source of the incivility or the victim's attribution of harmful intent. However, the conflict measure more clearly refers to harmful actions committed by other people, and the results suggest that if other people are identified as the source of stress, individuals are more likely to target people in return.

In addition to being a predictor of CWB, negative affectivity also appears to be an important moderator of the relationship between job stressors and CWB. Moderated regression analyses demonstrated that negative affectivity moderated the relationship between job stressors (workplace

incivility, interpersonal conflict, and organizational constraints) and CWB. In each case, the graphs of the moderators showed that the relationships between each of the job stressors and CWB was more strongly positive for individuals high in negative affectivity than for individuals low in negative affectivity. When low levels of incivility, conflict, or constraints were perceived, individuals appeared to engage in low levels of CWB overall. However, when high levels of stressors were reported, high negative affectivity individuals engaged in much higher levels of CWB than low negative affectivity individuals. Although the moderator results were mixed (i.e., none of the analyses using both peer- and self-report data were significant), they suggest that an individual's personality may impact how he or she responds job stress. Individuals high in negative affectivity, in particular, appear to use more counterproductive means to cope with job stressors. Perhaps with negative affectivity, it is the perception that matters most rather than the reality. Thus we find the moderator effect more with self-reported perceptions of incivility than peer reports.

While support was found for negative affectivity moderating the relationship between self-reported incivility and CWB, no significant moderator effect was found using peer-reported incivility as a predictor. Although some may be tempted to dismiss the significant moderator findings as artifacts of common method variance, Evans (1985) suggests that this is an unlikely explanation with moderator analyses. As one of the anonymous reviewers of this article commented, 'interaction effects are robust in the face of common method variance.' An alternative interpretation is that these results indicate that perspective is important. In other words, an individual's perception of the amount of incivility he/she experiences may be more important in predicting CWB than a seemingly more objective assessment of incivility such as peer-ratings. Peers may not witness all of the experiences of their co-workers and they may not perceive those experiences they do witness in the same way as their co-worker. According to Pearson et al. (2000), whether or not an act is deemed incivility depends upon the *interpretation* of the involved parties (p. 126). Therefore, that the results using self-reported incivility were more positive than those using peer-rated incivility is not necessarily surprising. Another possibility is lack of power. Researchers agree that moderator tests are low power (Aguinis, 1995; McClelland & Judd, 1993), therefore, given a sample size of 155 for peer data, a significant effect may have been harder to detect, especially if peer ratings of incivility are less accurate than self-reports.

The current study contained several limitations that may have affected the results. First, the use of undergraduate students as subjects has consistently been questioned. Although only students who were employed at the time of the study were allowed to participate and the results indicated that there was good variance in scores on all of the measures, employed students may differ in some meaningful way from other employed persons that may have biased their responses. For example, students may work in jobs they consider temporary, while they finish school and, thus, may behave differently than they would in jobs that have more career-relevance. However, Fox et al. (2001) contrasted employed students with non-student employees and found little difference in correlations of CWB with stressors (including conflict and constraints, emotions, and negative affectivity). This suggests that the use of students may not have posed a serious threat to the current findings. In fact, this population may be no more limiting than studies that use a single organization, since this sample was diverse in terms of both jobs and organizations.

In addition, the sample used in this study was predominantly female (only 23 per cent of subjects were male). Furthermore, because the participants in this study were asked to give a survey packet to one of their peers, the researcher had no control over who was chosen to participate. Participants may have given the surveys to individuals that they get along with fairly well or intentionally selected individuals who would make them look good instead of individuals who have the best opportunity to observe their behavior. Moreover, it is possible that not every participant gave the survey to a peer. If individuals who gave their peers the survey differed in some systematic way from those who did not give their peers the survey, then the results could be biased. Finally, as was mentioned previously,

the research design of the current study did not allow for causal conclusions to be drawn. More research is needed to establish the true directionality of the relationships between the variables explored in this study.

Despite these potential shortcomings, this study produced some interesting findings. Workplace incivility, a variable that has only recently received attention in organizational research, showed a similar pattern of relationships with job satisfaction and CWB as interpersonal conflict and organizational constraints, suggesting that it can also be considered a source of job stress. In general, experiencing workplace incivility and other job stressors appears to increase the likelihood that an individual will engage in CWB. Moreover, certain individuals, such as those high in NA, appear to be more likely to handle job stressors with less restraint.

Author biographies

Lisa M. Penney is an Assistant Professor of Industrial and Organizational Psychology at the University of Houston. She received her Ph.D. in I/O Psychology from the University of South Florida. Dr Penney's research interests include counterproductive behavior, social stressors at work, and organizational justice.

Paul E. Spector is a Professor of I/O psychology and the I/O doctoral program director at the University of South Florida. His work has appeared in many journals, including *Academy of Management Journal*, *Journal of Applied Psychology*, *Journal of Management*, *Journal of Organizational Behavior*, *Journal of Occupational and Organizational Psychology*, *Journal of Vocational Behavior*, *Organizational Behavior and Human Decision Processes*, *Personnel Psychology*, and *Psychological Bulletin*. At present he is the Point/Counterpoint editor for *Journal of Organizational Behavior*, and is on the editorial boards of *Journal of Occupational and Organizational Psychology*, *Organizational Research Methods*, and *Personnel Psychology*. In 1991 the Institute For Scientific Information listed him as one of the 50 highest impact contemporary researchers (out of over 102,000) in psychology worldwide.

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